

EMILY MACHUMA MUCHELE

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ADMINISTRATION OFFICER~FACILITIES MAINTENANCE OFFICER ~ ELECTRICAL ENGINEERING TECHNOLOGIST

Fostering Innovative solutions in administration and engineering operations to enhance compliance, safety and energy

Safe and efficient facility maintenance officer and engineering technologist with over 17 years' vast experience in all aspects of operations, facilities management, maintenance and repair. Having a sound knowledge of designing and evaluating a designs overall effectiveness, cost, reliability, and safety. Solution-oriented troubleshooter who cut costs and reduced error rates by improving operational effectiveness of mechanical systems. Leadership skills that have been demonstrated to drive optimization goals and improve organizational operations. Strong team member, able to motivate others to achieve optimal production rates while maintaining safety methods and practices. My goal is to become a valuable mutual asset by providing efficient engineering maintenance expertise contributing effectively to organizational success.

CORE COMPETENCIES

Process and Performance Improvements | Project Coordination | Maintenance Planning Strategies | Design Reinforcement Schemes | Computation of Quantities | Technical Writing | Contract Administration | Quality Assurance | Critical issue Resolution | Site Supervision I Administration services| Cost Analysis & Budgeting | Leadership and Mentorship | Electrical Building Systems | Special Projects

SELECTED MAJOR ACHIEVEMENTS

- **Administration Officer | Kenya School of Government**
 - ✓ Streamlined the operations in the department by introducing maintenance tracking records, improving request solving period from 80% to 95%.
 - ✓ Initiated acquisition of tablets for digital registration at the gate enhancing security services and traffic flow from averagely 1guest per 3 minute to 1 guest per minute.
 - ✓ Introduced daily casual workers staffing requirements and register improving on work output by 30%.
- **Maintenance Officer | Kenya School of Government**
 - ✓ Decreased maintenance costs by 60 % by employing preventative maintenance activities every month.
 - ✓ Reduced annual maintenance program costs from approximately (70% costs saving) plus to zero down time for the whole year.
 - ✓ Initiated energy efficiency program through energy audits (2015, 2022) which help identify areas of weakness and sealing energy leakages which reduced KPLC bill from an average of 2M to 1.3M per month.
 - ✓ Cultivated an extraordinary level of company loyalty, achieving a 10+ year tenure amongst 50% of employees.
 - ✓ Set up an in-house maintenance unit for the plants and vehicles which reduced equipment repairing and maintenance cost by 50%.

- **Electrical Technician/ Electrician**

- ✓ Performed electrical-related wiring and installation for residential buildings that were old and experienced electrical shocks occasionally, achieving 100% safety from shocks and fires.
- ✓ Consistently followed-up of service contracts of machines/equipment, submitted and followed-up electrical bill payments which achieved uninterrupted (99%) power supply in the school.
- ✓ Inspected and delivered 90% of electrical projects without faults saving the organization on maintenance service charges by 50%.
- ✓ Initiated use of maintenance logs and software (LAN SUPPORT) which cut response time delay by 50% increasing productivity and efficiency in the department by 70%, saving the school on maintenance costs by 20%.

EDUCATION & CREDENTIALS

- ✓ Master of Science in Renewable Energy Technology, Kenyatta University (Ongoing).
- ✓ Bachelor's degree in Electrical and Electronics, Technical University of Kenya.
- ✓ Higher Diploma in Electrical and Electronics Engineering, Technical University of Mombasa.
- ✓ Technician 3 (final) Certificate, Kenya Polytechnic University College.
- ✓ Craft 3 (final) Certificate, National Youth Service - NECTS.
- ✓ Grade 1 wire man Certificate in Electrical engineering and Electricians license.

CAREER HIGHLIGHTS

Administration Officer | Kenya School of Government | December 2024 to Date

Key Contributions and Results

- Overseeing daily activities and operations in Maintenance, Security, Transport and Grounds sections
- Guiding, directing, overseeing and supervising a team of technicians, drivers, grounds and security personnel for day-to-day operations.
- Supervising School's renovations and other minor projects in conjunction with procurement department to ensure service delivery.
- Managing day-to-day staffing and office requirements, planning and assigning work including establishing performance and development goals for the team.
- Preparing, filling and maintain monthly reports on the operations of maintenance, transport, security and grounds sections.
- Preparing department's memos, letters and documents for approval and signing of the Assistant Director, administration services.
- Identifying gaps in operational processes that inform on the new and effective ways of discharging duties across all sections in the department.
- Identifying, gathering and filling of performance Contract evidences.
- Representing the department in performance review meetings and evidence presentation.
- Raising requisitions for materials for various sections in the department

Maintenance Officer | Kenya School of Government | July 2015 to November 2024

Key Contributions & Results:

- Designing and planning basic electrical & mechanical systems including the selection of plant and equipment; project engineering and management of small-scale projects.
- Guiding, directing and overseeing a team of electrical & mechanical supervisors and technicians, including contract / annual maintenance for day-to-day operations and maintenance of electrical &

mechanical plant and equipment

- Providing technical advice to the designer, supervisor, inspector, and contracting service provider regarding installation and maintenance of facilities.
- Analyzing Mechanical, Electrical, and Plumbing engineering problems, carpentry, painting/decorations and environmental/gardening issues and formulate solutions.
- Identifying problems in the design, planning, management, and delivery of the projects in accordance with project requirements.
- Planning layout and installation of electrical wiring, equipment and fixtures and ensuring that the existing electrical, plumbing, carpentry and painting systems in the School are well maintained.
- Identifying and advising on any equipment that require maintenance contracts and recommending the best possible option.
- Supervising Schools renovations and other projects, and in conjunction with procurement department, advising on contracts and ensure timely completion of projects within budget.
- Managing day-to-day casual staffing requirements, planning and assigning work and establishing performance and development goals for team members.
- Providing mentoring, coaching and regular feedback to help manage conflict and improve team member performance.

Electrical Technician | Kenya School of Government | July 2012 - June 2015

Key Contributions & Results:

- Performed electrical installations, testing, inspection and maintenance of electrical system in all premises in the school and its residential houses.
- Initiated and carried out routine servicing/ maintenance of buildings, machinery and equipment.
- Diagnosed malfunctioning electrical system and components to locate causes of breakdowns and correct problems.
- Maintained optimal working conditions of all power systems (generator, UPSes, and Lifts) by establishing and enforcing preventive and maintenance and testing programs, scheduling repairs, coordinating shutdowns and installations.
- Supervised and coordinated between the School and Electrical Contractors to ensure quality and timely project completion, with Power Supplier to ensure uninterrupted power supply to the school.
- Ensured equipment defect reports were raised to facilitate timely repair and prepared reports after inspection of the equipment/Electrical Works.

Electrician | Kenya Institute of Administration | August 2008 - June 2012

Key Contributions & Results:

- Performed Electrical Installations, testing, inspection and maintenance of electrical systems in the School and its residential areas.
- Assessed and analyzed repairs and breakdown costs for equipment and recommended equipment upgrading or replacement.
- Took stock and prepared quarterly, semi-annual and annual Electrical budget in consolidation of Procurement officers.
- Conducted regular safety inspections and interpreted technical drawings as per work requirement.
- Repaired faulty electrical apparatus, components, and systems.
- Examined electrical apparatus, components, and systems to establish potential dangers.

Electrician | Edmas College & Unibilit 2003 Limited | January 2006 - July 2008

Key Contributions & Results:

- Implemented all electrical and electric systems and equipment installation, modification, repair and maintenance.
- Prepared and updated Electrical drawings during ongoing remodeling and renovations.
- Reviewed electrical plans and specifications in building in order to understand materials for electrical and electronics works.
- Changed, repaired and modified building electrical systems, wiring, fittings and appliances to suit end user interests.
- Performed scheduled general maintenance service on electrical wiring, systems, equipment and appliances.
- Implemented all electrical and electronics installation, maintenance and user requirements as specified by the manufacturer of systems and fixture.

PROFESSIONAL ROLES HELD

- ✓ Professional Member, Kenya Engineering Technology Registration Board (KETRB), August 2023 to date (Certificate No. PET00804/8-23, Serial No. 0001390)
- ✓ Member, Institute of Engineering Technologists (IET), April 2017 – To date (Certificate number-1153)
- ✓ Member, Inspection and Acceptance Committee, KSG – Jan. 2013 – December 2016
- ✓ Chairlady, Events Committee, KSG –January 2015- December 2018

PUBLICATION

Characterization of Food Wastes from the Hotel Industry as a Potential Feedstock for Energy Production.
<https://doi.org/10.1016/j.nexus.2025.100364>

REFERENCES

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